AQI Meter

About Myself

Ved Tiwari

- I am a highschool sophomore at Middletown High School.
- I have a lot of experience with programming mobile apps and building websites.
- My goal is to use my skills to make a positive difference in the world.

My Experience

- Before AQI Meter, I have built two apps and one website.
- My website can be found here: http://corona19stats.com/
- The website has received Delaware Senator recognition for engineering a user friendly, searchable website for local and global COVID-19 statistics.
- My google play account containing all of my previous mobile applications can be found here: https://play.google.com/store/apps/developer?id=Ved+Tiwari

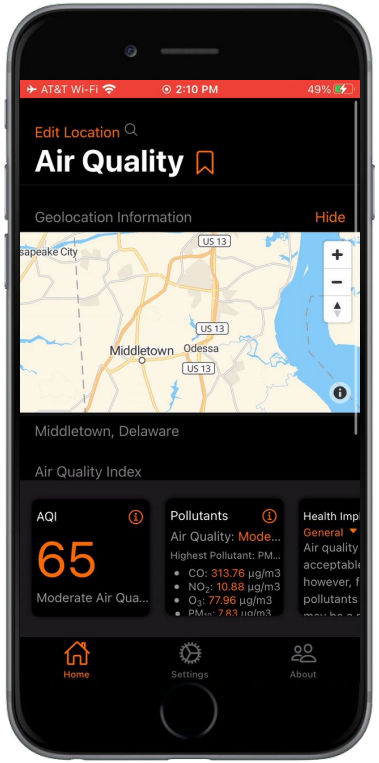
About AQI Meter

My Motivation Behind Building AQI Meter

As time progresses, our air quality continues to worsen at an ever alarming rate. Many factories and vehicles are emitting thousands of tons of emissions in the air and people who are living in places with more air pollution are suffering as a result. Myself and many others are concerned about the ongoing air quality issue. I was motivated to create this project because I noticed that there is not enough consolidated data that presents both pollutants and pollen markers in one app.

The information dashboards on AQI Meter allow users to understand healthy and unhealthy AQI Ranges and the sources of pollutants. I would like people to use AQI Meter and modify their actions based on the information to keep themselves safe.

AQI meter provides people of not only the AQI index, but also the individual pollutant rundown in their selected area. For those who have allergies in the springtime, AQI Meter also provides a breakdown of weed, grass, and tree pollen in the area. Additionally, AQI Meter also displays the physical location in the form of a map. The app provides air quality data for the user's current location upon opening the app.

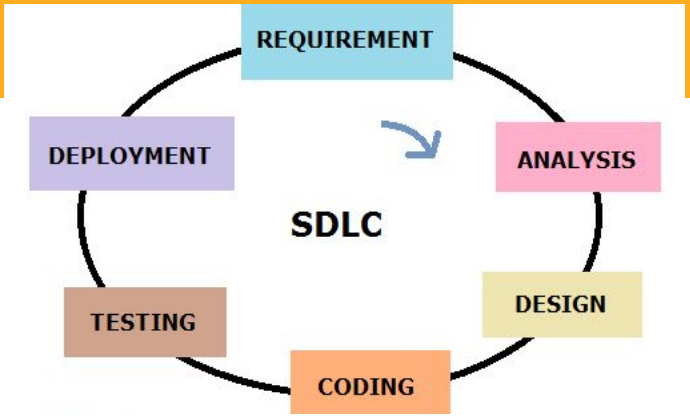


What Next

All of my websites and mobile applications have been aimed at one goal: to help people. Due to the large amount of data that AQI Meter offers, I would like to use this dataset in conjunction with other datasets and AI algorithms to use in machine learning. The data can easily be fed into such algorithms in order to predict future pollution levels, similar to the way the weather is forecasted in today's time.

How AQI Meter follows SDLC

Software Development Life Cycle



1.) Requirement

Before starting the project, I decided that I wanted to build an app which returns a detailed analysis of the air quality in the area. I also wanted the app to provide a 100% free experience for those who wanted information. Lastly, I also wanted the app to be fast and responsive, focusing on the user experience and convenience first.

2.) Analysis

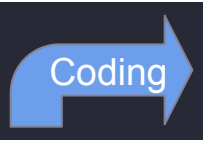
AQI Meter offers a suite of features which users can access at any time with no limitations. A few of these features includes the AQI for every city in the world followed by a pollutant by pollutant rundown of what is in the air at that moment. The tool also provides a geological preview of the area and surrounding cities in the form of a map. As a bonus, for those who are curious, AQI Meter also offers a weather forecast feature which displays the current temperature along with the humidity and pressure of the area.

3.) Design

For the design aspect of the AQI Meter, I wanted to make it look as mobile-friendly as possible. My goal was to make the app easy to navigate so users don't have to continually switch pages getting the information they want. I wanted all of the information on one page so users can access it at a swipe of a finger.

Early on in the development process, I made rough sketches on paper and whiteboard and I looked at examples from other mobile-friendly applications to get inspiration for my designs.

4.) Coding



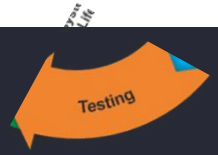
AQI Meter is written in JavaScript and uses many sources to retrieve information. These API's include:

1. OpenCage Data - Retrieves the latitude/longitude values of any city
2. Maptiler - Provides an interactive map for users to view
3. OpenWeatherMap - Provides AQI and weather data
4. WAQI - Provides the general status of the air at the current moment
5. ClimaCell - Retrieves Pollen data for tree, grass and weed pollen
6. EPA - Provides Health recommendations tuned to the AQI Rating

A large part of the design aspect of AQI Meter can be linked to the Bootstrap library for HTML, CSS and JavaScript.

The Bootstrap and jQuery libraries contributed to the aesthetic aspects of the project such as animations and icons

5.) Testing



Throughout the development process of AQI Meter, I performed system testing of the code in iterations to make sure that every part is ready for the user to view. During this process, I maintained a log to track bugs that appeared throughout the test cycles. I resolved all bugs before starting UAT testing.

During the UAT phase (user acceptance testing), I asked my friends and family to test the website and report any bugs, glitches, or recommendations to make AQI Meter the best as possible. After I received user input within the span of a week, I modified my code to satisfy these criteria.

5.) Deployment

Once the project was finished, it was time to package the files so they can be viewable through mobile and desktop devices.

For Android devices, I used the cordova package to convert my HTML and JavaScript files to a .APK file

For IOS devices, I decided to use PWA's (Progressive Web Application) and host the files to a web server. The user can visit such server to add the PWA to their home screen.

How to Download

Android

1.) Click this link to be redirected to the Google Play Store

https://play.google.com/store/apps/details?id=com.ved.app

2.) Click install to download AQI Meter

iPhone

1.) Visit this website on your iPhone

https://pocketagenda.org/

2.) Click the share button on the bottom of the screen

3.) Scroll to the bottom of the menu and click "add to homescreen"